# FINANCIAL TRANSACTION FRAUD DETECTION ANALYSIS

Detecting Fraud Patterns Through Data Analytics

**PRESENTED BY**
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01

# INTRODUCTION

## PROJECT OVERVIEW

Financial transaction fraud is a major challenge in digital banking and online payment systems.This project analyzes financial transaction data using Excel and Power BI to identify fraud patterns and generate actionable insights for better fraud detection.

## PROJECT OBJECTIVES

- Detect fraud patterns in transaction data.
- Perform data preprocessing & cleaning.
- Analyze trends using Pivot Tables.
- Develop an interactive Power BI Dashboard
- Generate actionable business insights.



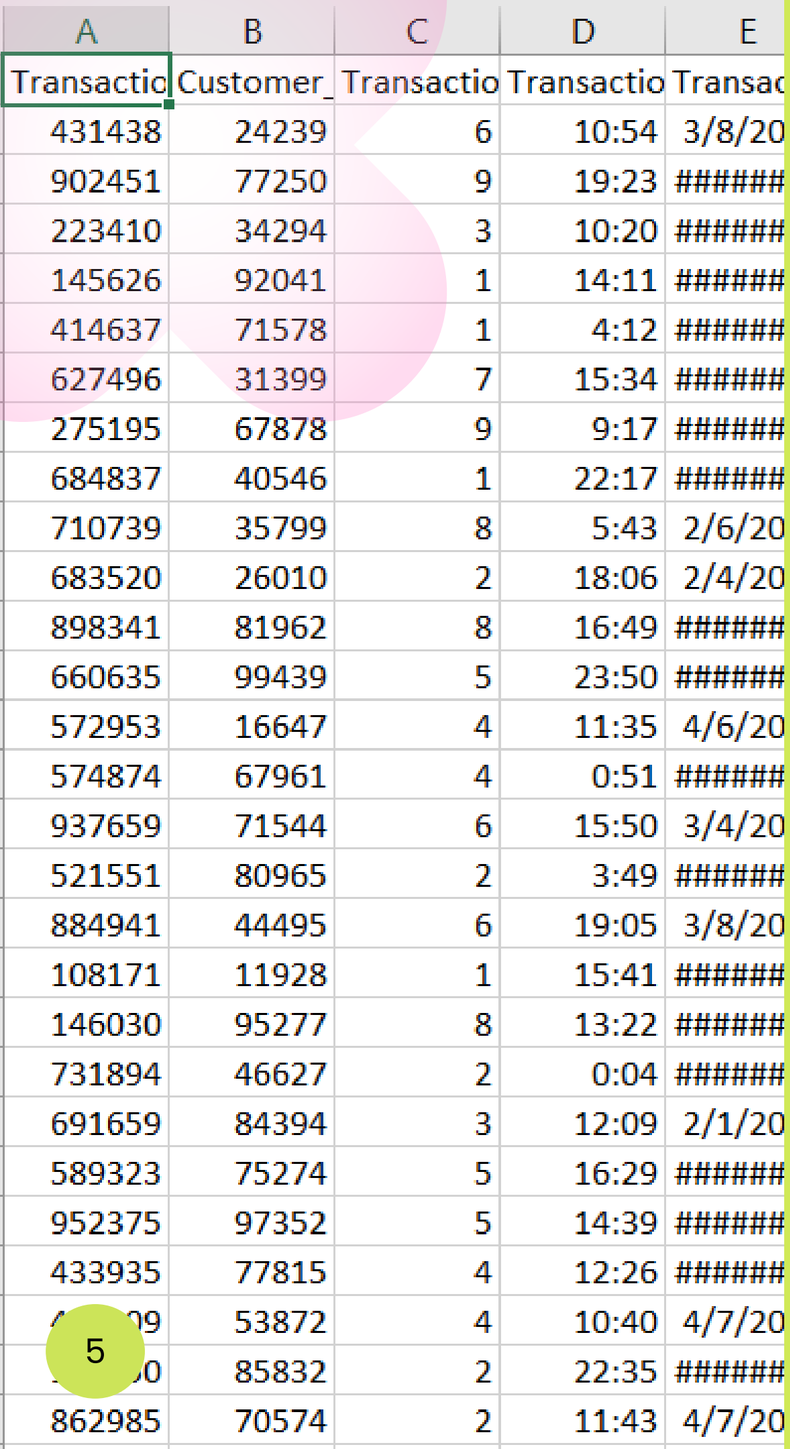
02

# DATASET OVERVIEW

Row_Dataset : https://www.kaggle.com/datasets/algozee/financial-transaction-fraud-dataset
Source : Kaggle
Features: 25
Total rows : 50001
Target : Fraud Label
Fraud Cases : 2422
Normal Cases : 47,527

## OBJECTIVE

Analyze transaction patterns, identify fraud indictors, and create interactive dashboards.



| A | B | C | D | E |
|---|---|---|---|---|
| Transactio | Customer_ | Transactio | Transactio | Transac |
| 431438 | 24239 | 6 | 10:54 | 3/8/20 |
| 902451 | 77250 | 9 | 19:23 | ###### |
| 223410 | 34294 | 3 | 10:20 | ###### |
| 145626 | 92041 | 1 | 14:11 | ###### |
| 414637 | 71578 | 1 | 4:12 | ###### |
| 627496 | 31399 | 7 | 15:34 | ###### |
| 275195 | 67878 | 9 | 9:17 | ###### |
| 684837 | 40546 | 1 | 22:17 | ###### |
| 710739 | 35799 | 8 | 5:43 | 2/6/20 |
| 683520 | 26010 | 2 | 18:06 | 2/4/20 |
| 898341 | 81962 | 8 | 16:49 | ###### |
| 660635 | 99439 | 5 | 23:50 | ###### |
| 572953 | 16647 | 4 | 11:35 | 4/6/20 |
| 574874 | 67961 | 4 | 0:51 | ###### |
| 937659 | 71544 | 6 | 15:50 | 3/4/20 |
| 521551 | 80965 | 2 | 3:49 | ###### |
| 884941 | 44495 | 6 | 19:05 | 3/8/20 |
| 108171 | 11928 | 1 | 15:41 | ###### |
| 146030 | 95277 | 8 | 13:22 | ###### |
| 731894 | 46627 | 2 | 0:04 | ###### |
| 691659 | 84394 | 3 | 12:09 | 2/1/20 |
| 589323 | 75274 | 5 | 16:29 | ###### |
| 952375 | 97352 | 5 | 14:39 | ###### |
| 433935 | 77815 | 4 | 12:26 | ###### |
| [illegible]09 | 53872 | 4 | 10:40 | 4/7/20 |
| [illegible]0 | 85832 | 2 | 22:35 | ###### |
| 862985 | 70574 | 2 | 11:43 | 4/7/20 |

# DATA PREPROCESSING

03

- Imported the row dataset
- Freeze pane → First row
- Converted to table format
- Centralized data
- Renamed columns
- Checked for missing values across all columns

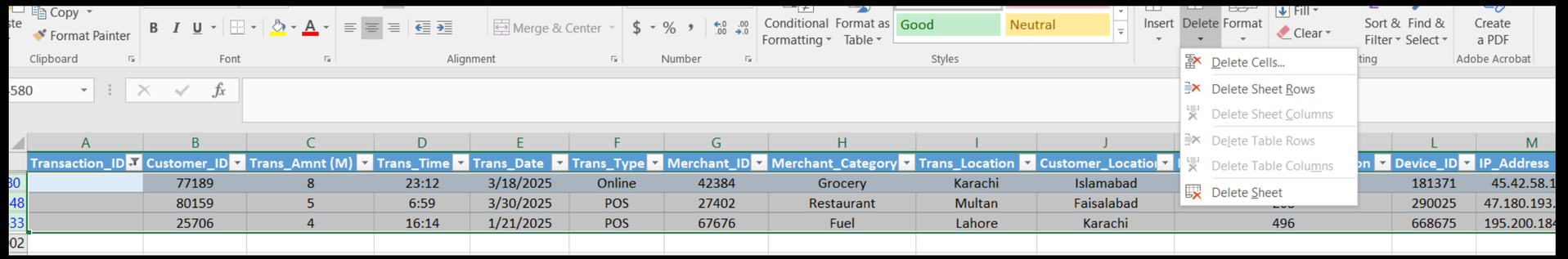
All categorical columns and numerical columns have blanks

- Handling missing values

| COLUMN | BLANK_ROW | ACTION |
|---|---|---|
| Transaction_ID | 3 | ID is a unique value and it cannot be meaningfully imputed.Deleted those rows. T_Rows : 49981 |
| Customer_ID | 10 | |
| Merchant_ID | 7 | |

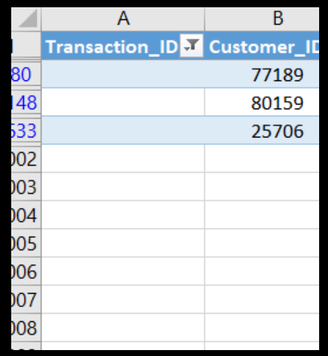
**Total Rows**

| | |
|---|---|
| 50000 | 444042 |
| 50001 | 121538 |
| 50002 | |

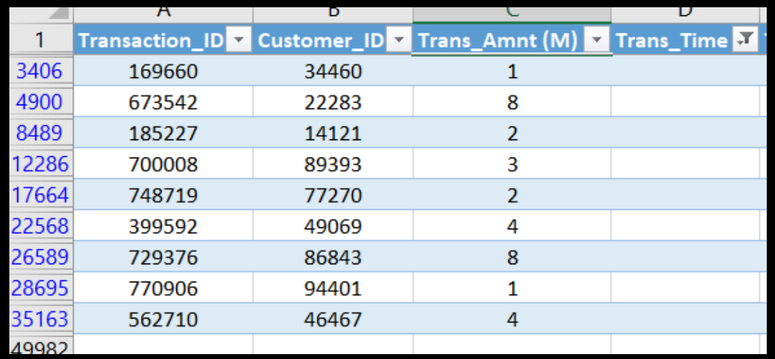



| Transaction_ID | Customer_ID | Trans_Amnt (M) | Trans_Time | Trans_Date | Trans_Type | Merchant_ID | Merchant_Category | Trans_Location | Customer_Location | Device_ID | IP_Address |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 77189 | 8 | 23:12 | 3/18/2025 | Online | 42384 | Grocery | Karachi | Islamabad | 181371 | 45.42.58.1 |
| | 80159 | 5 | 6:59 | 3/30/2025 | POS | 27402 | Restaurant | Multan | Faisalabad | 290025 | 47.180.193. |
| | 25706 | 4 | 16:14 | 1/21/2025 | POS | 67676 | Fuel | Lahore | Karachi | 668675 | 195.200.184 |

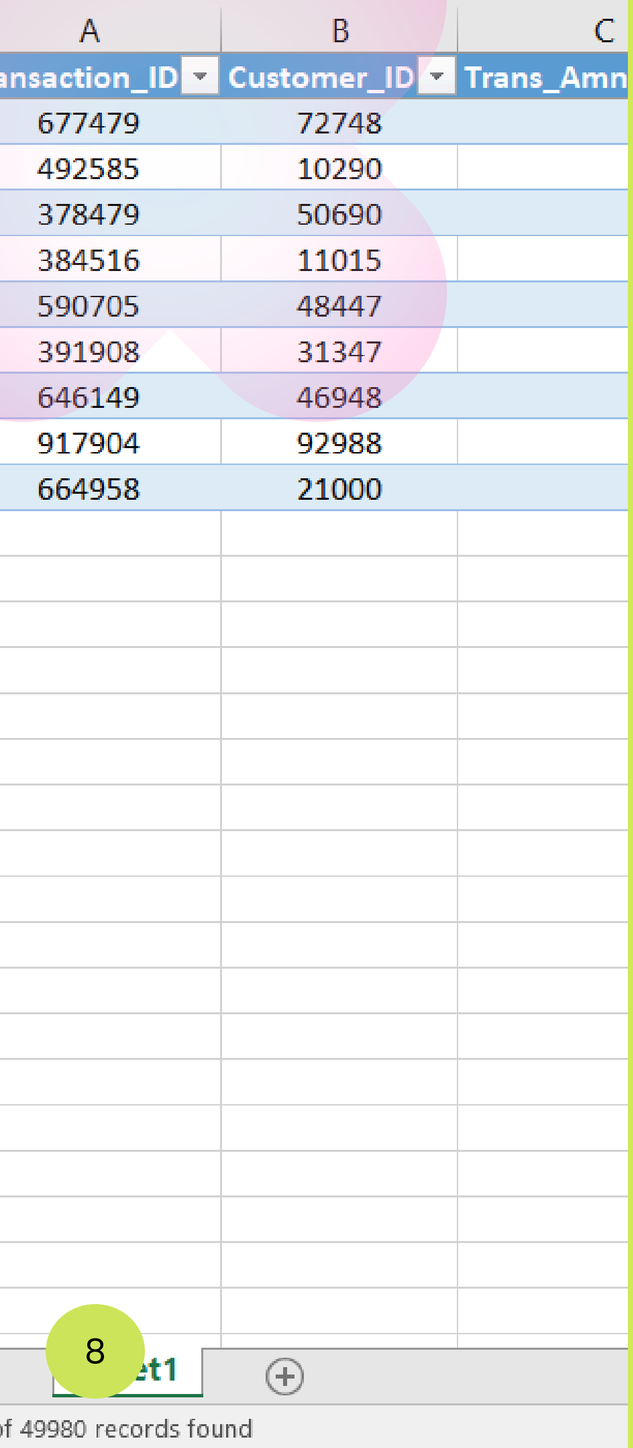
| COLUMN | BLANK ROWS | ACTION |
|---|---|---|
| Trans_Time<br>Trans_Date | 9<br>3 | Transaction time stamp represent actual event times,imputing with mode affect identifying unusual transaction.Therefore,deleted those rows. |
| Device_ID<br>IP_Address | 9<br>6 | Since these are unique identifier values cannot be imputed by median.Therefore,removed the affected rows. |
| Fraud_Label | 4 | The correct class could not be inferred without introducing bias,those records were removed to maintain the integrity . |



| | A | B |
|---|---|---|
| | Transaction_ID | Customer_ID |
| 80 | | 77189 |
| 148 | | 80159 |
| 633 | | 25706 |
| 002 | | |
| 003 | | |
| 004 | | |
| 005 | | |
| 006 | | |
| 007 | | |
| 008 | | |



| 1 | Transaction_ID | Customer_ID | Trans_Amnt (M) | Trans_Time |
|---|---|---|---|---|
| 3406 | 169660 | 34460 | 1 | |
| 4900 | 673542 | 22283 | 8 | |
| 8489 | 185227 | 14121 | 2 | |
| 12286 | 700008 | 89393 | 3 | |
| 17664 | 748719 | 77270 | 2 | |
| 22568 | 399592 | 49069 | 4 | |
| 26589 | 729376 | 86843 | 8 | |
| 28695 | 770906 | 94401 | 1 | |
| 35163 | 562710 | 46467 | 4 | |
| 49982 | | | | |



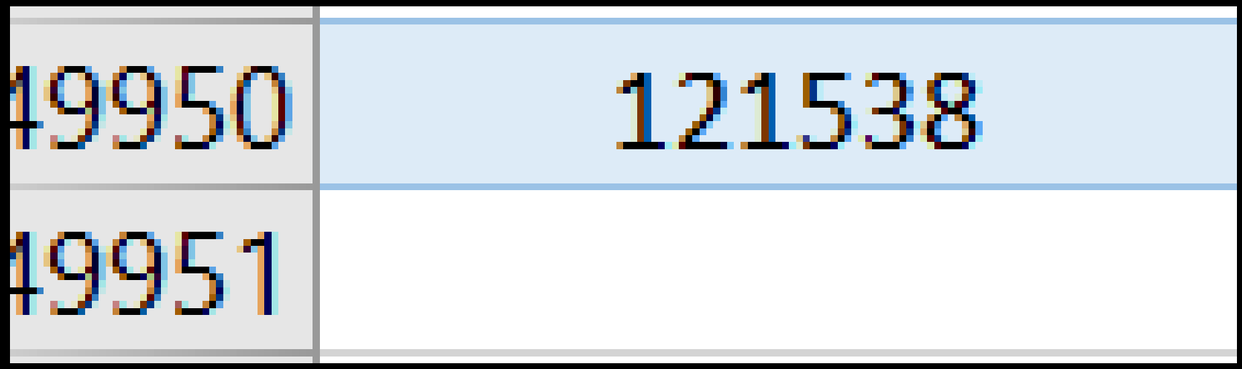
| A | B | C |
|---|---|---|
| ansaction_ID | Customer_ID | Trans_Amn |
| 677479 | 72748 | |
| 492585 | 10290 | |
| 378479 | 50690 | |
| 384516 | 11015 | |
| 590705 | 48447 | |
| 391908 | 31347 | |
| 646149 | 46948 | |
| 917904 | 92988 | |
| 664958 | 21000 | |



| COLUMN | BLANK ROWS | ACTION |
|---|---|---|
| Trans_Amount(M) | 9 | Since it can be influenced by outliers,making it more robust measure of central tendency replaced with median-5 =IF(ISBLANK(C2),MEDIAN(C:C),C2) |
| Trans_Type<br>Merchant_Category<br>Trans_Location<br>Customer_Location<br>Crad_Type | 3<br>9<br>6<br>4<br>3 | Created pivot tables for all corresponding columns to find the most occuring one and replace blanks with the mode-"Online","Restaurant","Multan","Lahore","Debit" |
| Dist_From_Customer_Location | 2 | Replaced with median-301 because some travel very less while others miles, this affect the mean value. Calculated median seperately and imputed to blank cells |
| Acc_Balance | 9 | Replaced blank with median-21 by calculating median of the customer account balance column seperately. |

| COLUMN | ROW BLANKS | ACTION |
|---|---|---|
| Daily_Trans_Count<br>Weekly_Trans_Count<br>Avg_Trans_Amount(M)<br>Max_Trans_Last_24h(M)<br>Failed_Trans_Count<br>Previous_Fraud_Count | 9<br>5<br>9<br>4<br>11<br>3 | These numerical columns are replaced with the median-”4",”13",”5",”1",”1" with respect to calulating median of each column seperately. |
| International_Trans_Count<br>New_Merchant<br>Unusual_Time_Trans | 4<br>5<br>3 | Blanks are imputed with mode-”Yes”,”Yes”,”No” by calculating values of each in pivot table |

**Total Rows After Handling Missing Values**



| Trans_Type | Count of Trans_Type |
|---|---|
| ATM | 16662 |
| Online | 16702 |
| POS | 16585 |
| **Grand Total** | **49949** |

| Trans_Location | Count of Trans_Location |
|---|---|
| Bangkok | 4984 |
| Dubai | 4911 |
| Faisalabad | 5012 |
| Islamabad | 5015 |
| Karachi | 4925 |
| Kuala Lumpur | 5064 |
| Lahore | 5044 |
| London | 4891 |
| Multan | 5082 |
| Singapore | 5021 |
| **Grand Total** | **49949** |

| Cust_Location | Count of Customer_Location |
|---|---|
| Faisalabad | 9894 |
| Islamabad | 10005 |
| Karachi | 10000 |
| Lahore | 10111 |
| Multan | 9939 |
| **Grand Total** | **49949** |

| Count of Trans | Trans_Amount_C | | | |
|---|---|---|---|---|
| Merchant_ | Low | Medium | High | Grand Tota |
| Electronics | 1894 | 3638 | 2679 | 8211 |
| Grocery | 1793 | 3698 | 2751 | 8242 |
| Fuel | 1845 | 3746 | 2757 | 8348 |
| Clothing | 1839 | 3675 | 2761 | 8275 |
| ATM | 1824 | 3736 | 2833 | 8393 |
| Restaurant | 1885 | 3725 | 2870 | 8480 |
| **Grand Total** | **11080** | **22218** | **16651** | **49949** |

| Merchant_Category | Count of Merchant_Category |
|---|---|
| ATM | 8393 |
| Clothing | 8275 |
| Electronics | 8211 |
| Fuel | 8348 |
| Grocery | 8242 |
| Restaurant | 8480 |
| **Grand Total** | **49949** |

| Usual_Time | Count of Unusual_Time_Trans |
|---|---|
| No | 25088 |
| Yes | 24862 |
| (blank) | |
| **Grand Total** | **49950** |

| Trans_Day | Count of Trans_Day |
|---|---|
| Sunday | 6921 |
| Tuesday | 6972 |
| Monday | 7004 |
| Saturday | 7021 |
| Friday | 7151 |
| Wednesday | 7385 |
| Thursday | 7495 |
| **Grand Total** | **49949** |

| International_Trans | Count of International_Trans |
|---|---|
| No | 24906 |
| Yes | 25043 |
| **Grand Total** | **49949** |

| Card_Type | Count of Card_Type |
|---|---|
| Credit | 24869 |
| Debit | 25080 |
| **Grand Total** | **49949** |

| New_Merchant | Count of New_Merchant |
|---|---|
| No | 24914 |
| Yes | 25035 |

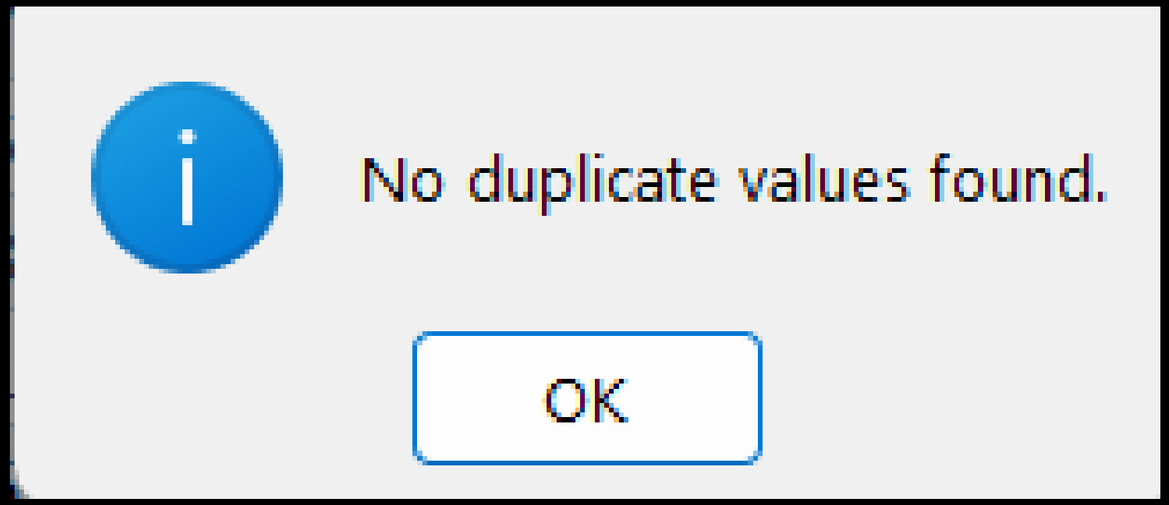
| | |
|---|---|
| avg | 5.00004 |
| median | 5 |
| avg | 300.1271 |
| median | 301 |
| avg | 20.99684 |
| median | 21 |
| median | 4 |
| median | 13 |
| median | 3 |
| median | 5 |
| median | 1 |
| median | 1 |

$fx$ =IF(ISBLANK([@[Trans_Amnt (M)]]),MEDIAN([Trans_Amnt (M)]),[@[Trans_Amnt (M)]])

| | C | D | E | F | G | |
|---|---|---|---|---|---|---|
| _ID | Trans_Amnt (M) | Column1 | Trans_Time | Trans_Date | Trans_Type | Mercha |
| 9 | 6 | 6 | 10:54 | 3/8/2025 | POS | 97 |
| 0 | 9 | 9 | 19:23 | 1/17/2025 | ATM | 27 |

- Duplicate Handling

Select the whol datsaset → Data → Remove Duplicates → No Duplicates Found




- Outlier Detection

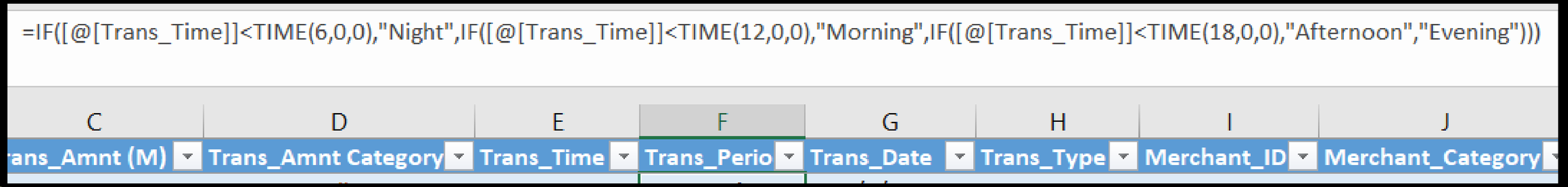
Outliers occurs due to data entry error, genuine unusual event and frauds. Method used here to find the outlier is IQR method. It measures spread of middle 50% of the data. Since all numerical values fall within their respective IQR limits,no statistical outliers were identified.

Column Outliers

| COLUMN | Q1 | MEDIAN | Q3 | IQR | L.B | H.B |
|---|---|---|---|---|---|---|
| Acc_Balance(M) | 12 | 21 | 30 | 18 | -15 | 57 |
| Daily_Trans_Count | 2 | 4 | 6 | 4 | -4 | 12 |
| Weekly_Trans_Count | 7 | 13 | 18 | 11 | -9.5 | 34.5 |
| Avg_Trans_Amount(M) | 2 | 2 | 4 | 2 | -1 | 7 |
| Max_Trans_Last_24h(M) | 3 | 5 | 7 | 4 | -3 | 13 |
| Failed_Trans_Count | 0 | 1 | 2 | 2 | -3 | 5 |
| Prev_Fraud_Count | 0 | 1 | 1 | 1 | -1.5 | 2.5 |

- Excel Analysis-Feature Engineering

| COLUMN | FORMULA IMPUTED |
|---|---|
| Trans_Amount_Category | =IF([@[Trans_Amnt (M)]]<3,"Low",IF([@[Trans_Amnt (M)]]<=6,"Medium","High")) |
| Time_Period | =IF([@[Trans_Time]]<TIME(6,0,0),"Night",IF([@[Trans_Time]]<TIME(12,0,0),"Morning",IF([@[Trans_Time]]<TIME(18,0,0),"Afternoon","Evening"))) |
| Trans_Day | =TEXT([@[Trans_Date]],"dddd") |
| Merch_Category_Risk_Level | =IF(OR([@[Merchant_Category]]="ATM",[@[Merchant_Category]]="Restaurant"),3,IF(OR([@[Merchant_Category]]="Grocery",[@[Merchant_Category]]="Clothing",[@[Merchant_Category]]="Fuel"),2,1)) |



- Conditional Formatting

1. Highlighted the high risk Merchant_Categories → "ATM & Restaurant" by "Red Flag", medium risk Merchant_Categories → "Grocery,Fuel & Clothing by "Yellow Flag", and low risk Merchant_Category → "Electronics" by "Green Flag" in Merchant_Risk_Category_Level Column. Risk level is categorised with the help pivot values we derived.

2. Highlighted most occuring Trans_Type → "Online" in "Green color".

3. Highlighted most used transaction CArd_Type → "Debit" in "Green color".

4. Highlighted the Fraud_Label → "Fraud" in "Red color" and "Normal" in "Green color".

5. Highlighted the Trans_Day → Wednesday, Thursday, and Friday with Red color as the count of transaction is higher these days so probability of fraud occurance is high.

6. Highlighted the Trans_Amount(M) → "less than 3" in "Green", "less than or equal to 6" in "Orange", and "greater than 9" in Red with respect to the Trans_Amount_Category column.

| 1 | Transacti | Custome | Trans_Amo | Trans_Amt C | Trans_ | Trans_P | Trans_ | Trans_ | Trans_ | Merchan | Merchant_C | M | Trans_Lo | Customer_Lo | Dist_From_Custome | Device | IP_Addr | Card_ | Acc_Balan | Daily_Trans | Weekly_Trans | Avg_Trans_A | Max_Trans_Last | International | New_Merch | Failed_Trans | Unusual_Time | Previous_Frau | Fraud_ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 17 | 521551 | 80365 | 2 | Low | 3:43 | Night | 1/28/2025 | Tuesday | ATM | 41227 | ATM | | Singapore | Multan | 308 | 326450 | 123.114.35.36 | Credit | 11 | 6 | 18 | 3 | 2 | Yes | Yes | 2 | Yes | 1 | Normal |
| 18 | 864341 | 44435 | 6 | Medium | 13:05 | Evening | 3/8/2025 | Saturday | POS | 62567 | ATM | | Islamabad | Karachi | 568 | 436703 | 33.172.25.33 | Debit | 20 | 3 | 2 | 3 | 8 | No | No | 1 | No | 0 | Normal |
| 19 | 108171 | 11328 | 1 | Low | 15:41 | Afternoon | 1/10/2025 | Friday | Online | 42828 | Restaurant | | Kuala Lumpur | Faisalabad | 10 | 368757 | 37.244.104.24 | Debit | 18 | 6 | 17 | 5 | 1 | No | Yes | 2 | Yes | 0 | Normal |
| 20 | 146030 | 35277 | 8 | High | 13:22 | Afternoon | 4/23/2025 | Wednesday | Online | 51759 | Fuel | | Kuala Lumpur | Lahore | 338 | 885330 | 37.155.102.151 | Credit | 18 | 2 | 16 | 5 | 5 | Yes | No | 2 | No | 1 | Normal |
| 21 | 731834 | 46627 | 2 | Low | 0:04 | Night | 4/24/2025 | Thursday | POS | 27901 | Electronics | | Singapore | Karachi | 181 | 308400 | 253.17.246.211 | Credit | 21 | 7 | 3 | 5 | 5 | No | Yes | 2 | Yes | 1 | Normal |
| 22 | 691659 | 84394 | 3 | Medium | 12:09 | Afternoon | 2/1/2025 | Saturday | ATM | 69389 | Grocery | | Islamabad | Multan | 174 | 506553 | 36.136.124.20 | Credit | 20 | 2 | 16 | 2 | 6 | Yes | No | 0 | No | 0 | Normal |
| 23 | 589323 | 75274 | 5 | Medium | 16:29 | Afternoon | 4/18/2025 | Friday | ATM | 64305 | Electronics | | Singapore | Karachi | 565 | 686953 | 35.254.23.80 | Debit | 3 | 7 | 24 | 5 | 6 | Yes | Yes | 1 | Yes | 1 | Normal |
| 24 | 352315 | 97352 | 5 | Medium | 14:39 | Afternoon | 3/15/2025 | Saturday | POS | 21221 | Restaurant | | Kuala Lumpur | Multan | 49 | 184424 | 11.73.89.39 | Credit | 35 | 4 | 21 | 3 | 7 | Yes | Yes | 1 | No | 0 | Normal |
| 25 | 433935 | 77815 | 4 | Medium | 12:26 | Afternoon | 1/11/2025 | Saturday | ATM | 38082 | Restaurant | | Lahore | Faisalabad | 322 | 867596 | 53.187.178.186 | Debit | 37 | 4 | 6 | 1 | 1 | Yes | Yes | 1 | No | 1 | Normal |
| 26 | 472909 | 53872 | 4 | Medium | 10:40 | Morning | 4/7/2025 | Monday | Online | 40537 | Grocery | | Dubai | Faisalabad | 83 | 460100 | 71.3.65.233 | Credit | 20 | 4 | 13 | 5 | 4 | Yes | Yes | 2 | No | 1 | Normal |
| 27 | 952450 | 85832 | 2 | Low | 22:35 | Evening | 2/22/2025 | Saturday | ATM | 71995 | Restaurant | | Bangkok | Islamabad | 531 | 751144 | 214.102.90.38 | Debit | 15 | 3 | 23 | 5 | 5 | No | No | 1 | No | 1 | Normal |
| 28 | 862385 | 70574 | 2 | Low | 11:43 | Morning | 4/7/2025 | Monday | POS | 33977 | Clothing | | Kuala Lumpur | Lahore | 221 | 340393 | 45.60.238.218 | Debit | 28 | 7 | 2 | 5 | 2 | No | Yes | 1 | No | 0 | Normal |
| 29 | 275707 | 92510 | 9 | High | 2:53 | Night | 1/8/2025 | Wednesday | POS | 83227 | Fuel | | Islamabad | Lahore | 436 | 350307 | 14.146.172.24 | Credit | 26 | 6 | 15 | 1 | 7 | Yes | Yes | 1 | No | 1 | Normal |
| 30 | 774390 | 86133 | 6 | Medium | 23:49 | Evening | 1/24/2025 | Friday | Online | 10160 | Restaurant | | Faisalabad | Islamabad | 215 | 672046 | 246.237.215.161 | Debit | 24 | 6 | 13 | 5 | 7 | Yes | No | 1 | Yes | 0 | Normal |
| 31 | 223921 | 19832 | 1 | Low | 10:53 | Morning | 4/23/2025 | Tuesday | POS | 17632 | Electronics | | Multan | Faisalabad | 431 | 178320 | 3.65.33.193 | Debit | 11 | 4 | 10 | 3 | 4 | Yes | No | 0 | No | 1 | Normal |
| 32 | 130850 | 73966 | 4 | Medium | 1:21 | Night | 4/12/2025 | Saturday | ATM | 12323 | Grocery | | Islamabad | Islamabad | 436 | 166738 | 73.216.212.67 | Debit | 15 | 7 | 16 | 1 | 6 | Yes | No | 2 | Yes | 0 | Normal |
| 33 | 436203 | 48635 | 9 | High | 15:08 | Afternoon | 3/31/2025 | Monday | POS | 37532 | ATM | | Dubai | Lahore | 330 | 946301 | 37.135.15.111 | Credit | 31 | 4 | 18 | 3 | 5 | Yes | Yes | 1 | No | 1 | Normal |
| 34 | 301516 | 12183 | 9 | High | 11:27 | Morning | 3/21/2025 | Friday | Online | 87513 | ATM | | Kuala Lumpur | Lahore | 151 | 835781 | 119.120.228.20 | Debit | 36 | 3 | 20 | 1 | 3 | Yes | No | 1 | Yes | 1 | Normal |
| 35 | 568626 | 73233 | 7 | High | 7:49 | Morning | 2/8/2025 | Saturday | ATM | 82648 | Electronics | | Dubai | Karachi | 431 | 696580 | 125.251.6.8 | Credit | 32 | 6 | 12 | 2 | 1 | No | No | 1 | Yes | 1 | Normal |

04

- **EXPLORATORY DATA ANALYSIS (EDA)**

PURPOSE OF EDA

Understand the structure of dataset, Identify transaction patterns, Detect unusual behaviour and potential analysis, Prepare insights for dashboard development.

KEY AREAS EXPLORED

Transaction Amount, Fraud Distribution, Merchant Categories, Customer Locations, Time Based Trends, and Payment Channels .

STATISTICAL SUMMARY

| METRIC | VALUE |
|---|---|
| Total Records | 49,949 |
| Total Columns | 29 |
| Numerical Columns | 10 |
| Categorical Columns | 12 |
| Identifier Columns | 5 |
| Date & Time Columns | 2 |
| Missing Values | 0 |
| Duplicate Records | 0 |
| Fraud Transaction | 2422 |
| Normal Transaction | 47,527 |
| Fraud Percentage | 5% |

## KEY OBSERVATIONS

- Majority of transactions are legitimate.
- Fraud case represent upto 5% of the dataset.
- Dataset is imbalanced.
- Suitable for Fraud Detection Analysis

## VISUAL EXPLORATION

1. Transaction distribution by Time Period
2. Transaction Volume by Transaction Type
3. Fraud Occurance by Failed Transaction
4. Fraud Analysis for New Merchant vs Fraud
5. Frau Distribution Across Customer Location
6. Fraud Analysis by Merchant Category
7. Slicer → Transaction Day
8. Slicer → Card Type

## EDA outcome

Patterns identified during EDA were later validated using Pivot Tables and Power BI dashboards.

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- **EXCEL ANALYSIS-PIVOT TABLES**

1. Transaction distribution by Time Period

2.Transaction Volume by Transaction Type

3.Fraud Occurance by Failed Transaction

4.Fraud Analysis for New Merchant vs Fraud

5.Frau Distribution Across Customer Location

6.Fraud Analysis by Merchant Category

7.Slicer → Transaction Day

8.Slicer → Card Type

# Pivot tables created

| TRANSACTION DISTRIBUTION by TIME PERIOD | |
|---|---|
| Trans_Period | Count of Transaction_ID |
| Afternoon | 12570 |
| Morning | 12503 |
| Night | 12439 |
| Evening | 12437 |
| Grand Total | 49949 |

Trans_Day

Sunday

Monday

Tuesday

Wednesday

| TRANSACION VOLUME by TRANSACTION TYPE | |
|---|---|
| Trans_Type | Sum of Trans_Amnt (M) |
| ATM | 83298.00 |
| Online | 83601.00 |
| POS | 82848.00 |
| Grand Total | 249747.00 |

Card_Type

Credit

Debit

| FRAUD OCCURANCE by FAILED TRANSACTION | | | |
|---|---|---|---|
| Count of Transaction_ID | Fraud_Label | | |
| Failed _Trans | Fraud | Normal | Grand Total |
| 0 | 803 | 16011 | 16814 |
| 1 | 847 | 15841 | 16688 |
| 2 | 772 | 15675 | 16447 |
| Grand Total | 2422 | 47527 | 49949 |

| FRAUD ANALYSIS FOR NEW MERCHANT vs FRAUD | | | |
|---|---|---|---|
| Count of Transaction_ID | Fraud_Label | | |
| New_Merchant | Fraud | Normal | Grand Total |
| No | 1197 | 23717 | 24914 |
| Yes | 1225 | 23810 | 25035 |
| Grand Total | 2422 | 47527 | 49949 |

| FRAUD DISTRIBUTION ACROSS CUSTOMER LOCATION | | | |
|---|---|---|---|
| Count of Trans_Amnt (M) | Fraud_Label | | |
| Cust_Location | Fraud | Normal | Grand Total |
| Faisalabad | 480 | 9414 | 9894 |
| Islamabad | 515 | 9490 | 10005 |
| Karachi | 464 | 9536 | 10000 |
| Lahore | 476 | 9635 | 10111 |
| Multan | 487 | 9452 | 9939 |
| Grand Total | 2422 | 47527 | 49949 |

| FRAUD ANALYSIS by MERCHANT CATEGORY | | | |
|---|---|---|---|
| Count of Transaction_ID | Fraud_Label | | |
| Merchant_Category | Fraud | Normal | Grand Total |
| ATM | 420 | 7973 | 8393 |
| Clothing | 380 | 7895 | 8275 |
| Electronics | 389 | 7822 | 8211 |
| Fuel | 408 | 7940 | 8348 |
| Grocery | 397 | 7845 | 8242 |
| Restaurant | 428 | 8052 | 8480 |
| Grand Total | 2422 | 47527 | 49949 |

05

- Loaded the data into **Power BI** & transformed data into Power Query
- Changed the datatypes of Transaction_ID, Customer_ID, Merchant_ID, Device_ID into text because these are unique features.
- Closed & applied.
- Created **KPI Measures** :

1. Total_Transaction
   Total_Transactions = COUNT(Table1[Transaction_ID])
2. Total Transaction Amount
   Total_Trans_Amnt = SUM(Table1[Trans_Amnt (M)])
3. Total Normal Transaction
   Total_Normal_Trans = CALCULATE(COUNT(Table1[Transaction_ID]),Table1[Fraud_Label]="Normal")
4. Total Fraud Transaction
   Total_Fraud_Trans = CALCULATE(COUNT(Table1[Transaction_ID]),Table1[Fraud_Label]="Fraud")
5. Total Customers
   Total_Customers = DISTINCTCOUNT(Table1[Customer_ID])
6. Total Fraud Transaction Amount
   Fraud_Trans_Amnt = CALCULATE(SUM(Table1[Trans_Amnt (M)]),Table1[Fraud_Label]="Fraud")
7. Fraud Rate Percentage
   Fraud_Rate% = DIVIDE([Total_Fraud_Trans],[Total_Transactions],0)
8. Total Merchant
   Total_Merchants = DISTINCTCOUNT(Table1[Merchant_ID])
9. Average Transaction Amount
   Avg_Trans_Amnt = AVERAGE(Table1[Trans_Amnt (M)])

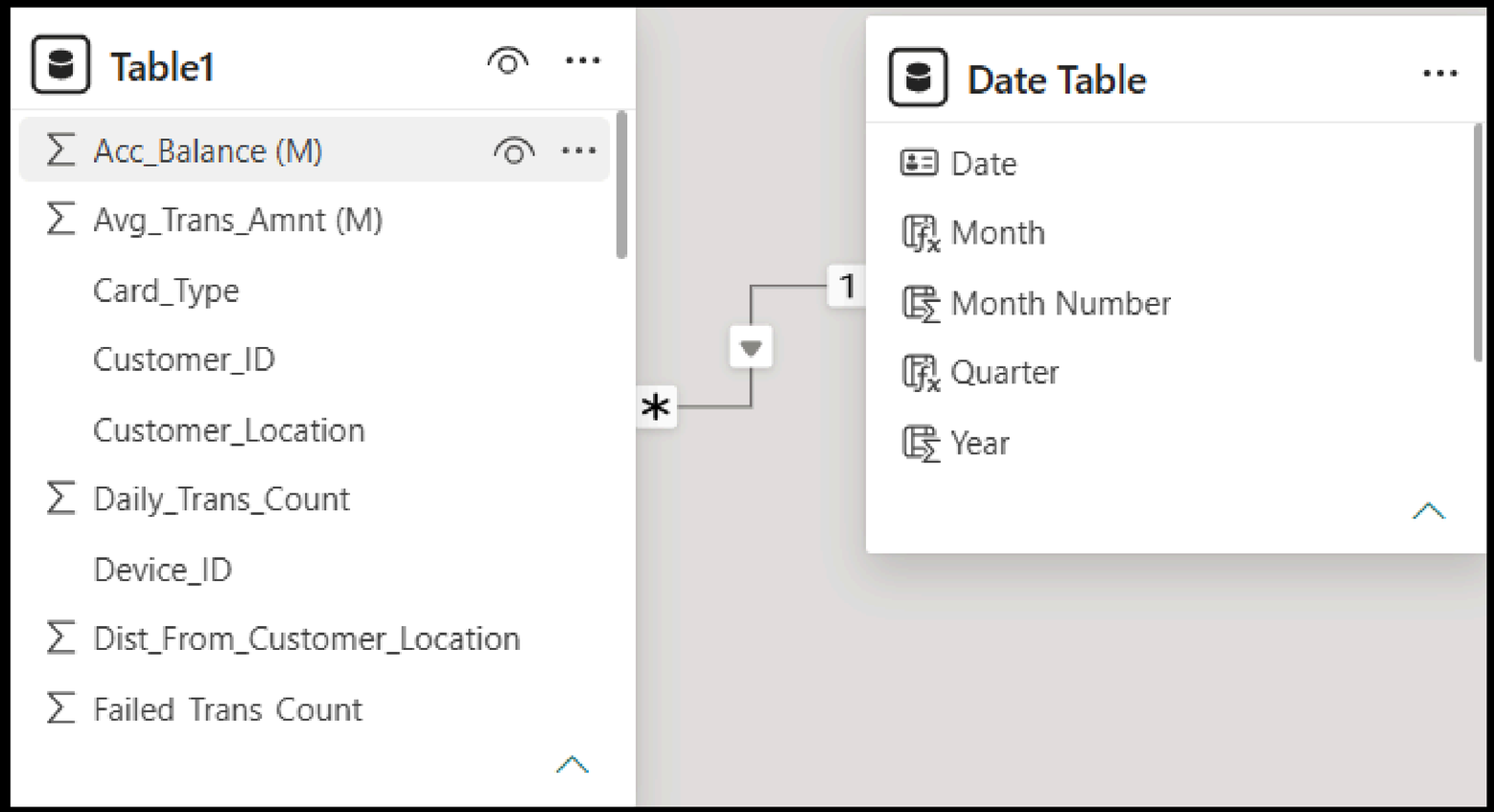
- Created the Date Table using DAX Function

  Date Table = CALENDAR(MIN(Table1[Trans_Date]),MAX(Table1[Trans_Date]) )

- Extended the table by adding the coulmns

  1. Year Column
     Year = YEAR('Date Table'[Date])
  2. Month Column
     Month = FORMAT('Date Table'[Date],"MMMM")
  3. Year-Month Column
     Year-Month = FORMAT('Date Table'[Date],"MMM YYYY")
  4. Month Number Column
     Month Number = MONTH('Date Table'[Date])
  5. Quarter Column
     Quarter = "Q" & QUARTER('Date Table'[Date])
  6. Date Column
     Date Table = CALENDAR(MIN(Table1[Trans_Date]),MAX(Table1[Trans_Date]) )

- Established relationship in the Model View




- Created visuals for the Dashboard
- Arranged visuals in two pages where first page shows the overall Fraud Detection in Banking and KPI Cards and the second page shows the deeper diving in to fraud analysis through diiferent categories and have slicers .

- The charts created are :

  1. Count of Transaction ID vs Fraud Label → Donut Chart
  2. Fraud Trend vs Fraud Rate % → Line & Stacked Column Chart
  3. Fraud Transaction Amount by Time Period → Stacked Column Chart
  4. Total Failed Transaction by Failed Transaction Count → Stacked Column Chart
  5. Fraud Distribution by Customer Location → Map Visual
  6. Fraud Transaction by Merchant Category → Stacked Bar Chart
  7. Merchant Category Fraud Label → Treemap Visual
  8. Fraud Transaction by Customer Location → Stacked Bar Chart
  9. Fraud Across Transaction Types → Clustered Column Chart

- KPI cards created are :

  1. Total Transaction Count
  2. Total Transaction Amount
  3. Total Fraud Transaction Count
  4. Total Fraud %
  5. Count of Total Customers
  6. Count of Total Merchants

- Slicers added are :

1. Fraud Label
2. Merchant Category
3. Customer Location
4. Date
5. Transaction Type

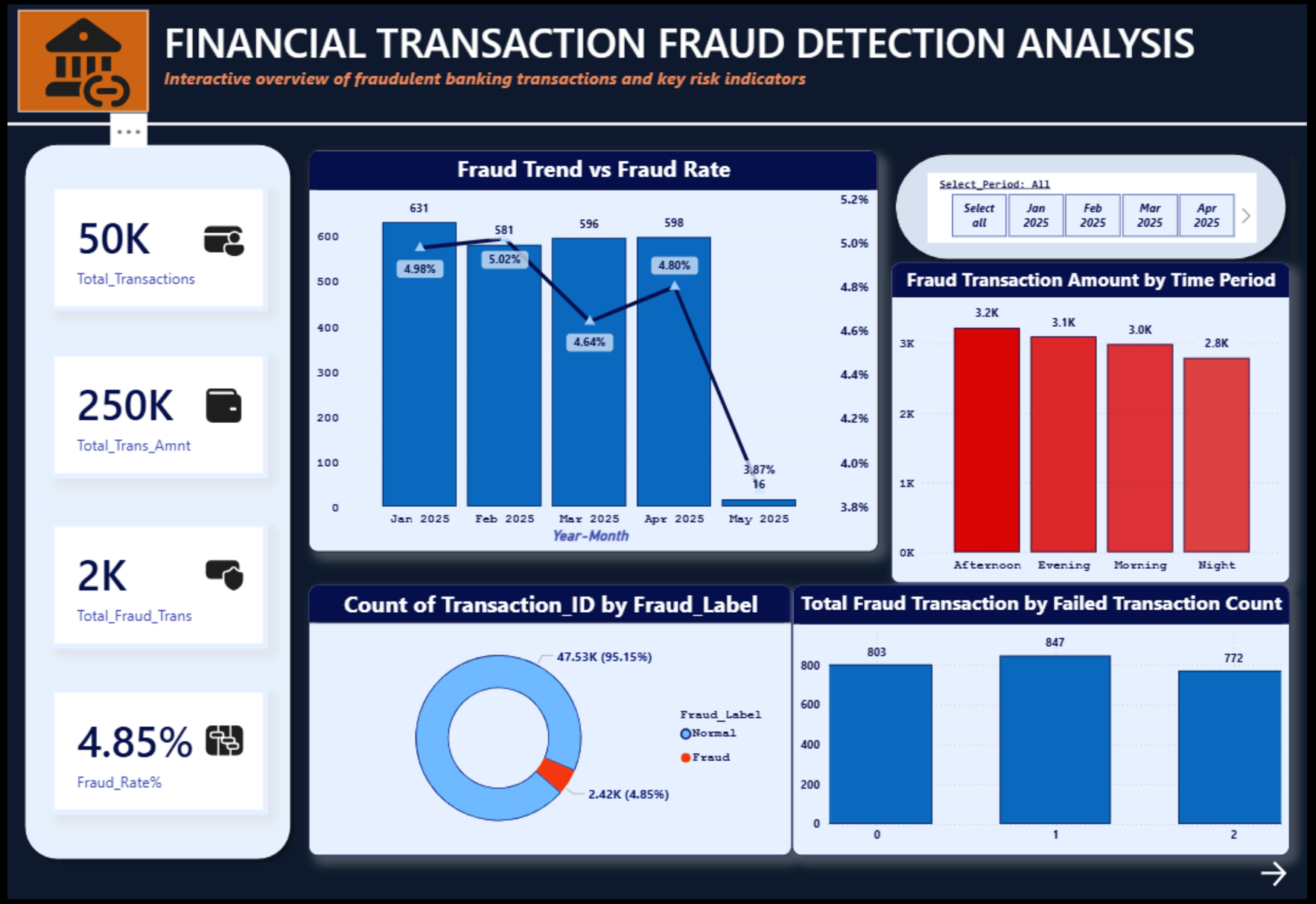
- Polished and edited the Visuals, Slicers and KPI’s to get two interactive Dashboards.

# KEY INSIGHTS

07

- Fraud transactions are distributed across all customer locations.
- ATM, Online, POS transactions contain fraudulent activities.
- Merchant categories shows varying  level of fraud occurence.
- Interactive slicers enable quick fraud analysis by location, transaction type, merchant catewgory and fraud status.
- The  Dashboards helps identify fraud patterns and supports faster decision-making.

# DASHBOARD


FINANCIAL TRANSACTION FRAUD DETECTION ANALYSIS
Interactive overview of fraudulent banking transactions and key risk indicators
50K
Total_Transactions
250K
Total_Trans_Amnt
2K
Total_Fraud_Trans
4.85%
Fraud_Rate%
Fraud Trend vs Fraud Rate
631
581
596
598
16
4.98%
5.02%
4.64%
4.80%
3.87%
Jan 2025
Feb 2025
Mar 2025
Apr 2025
May 2025
Year-Month
Select_Period: All
Select all
Jan 2025
Feb 2025
Mar 2025
Apr 2025
Fraud Transaction Amount by Time Period
3.2K
3.1K
3.0K
2.8K
Afternoon
Evening
Morning
Night
Count of Transaction_ID by Fraud_Label
47.53K (95.15%)
2.42K (4.85%)
Fraud_Label
Normal
Fraud
Total Fraud Transaction by Failed Transaction Count
803
847
772

# FINANCIAL TRANSACTION FRAUD DETECTION ANALYSIS

*Interactive overview of fraudulent banking transactions and key risk indicators*

**FRAUD LABEL**

- ☐ Fraud
- ☐ Normal

**TRANSACTION TYPE**

- ☐ ATM
- ☐ Online
- ☐ POS

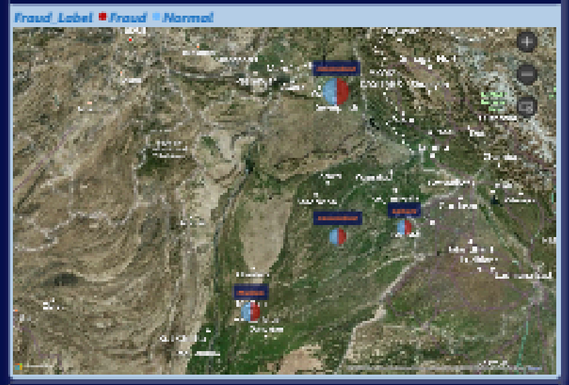
**CUSTOMER LOCATION**

| | |
|---|---|
| Select all | Karachi |
| Faisalabad | Lahore |
| Islamabad | Multan |

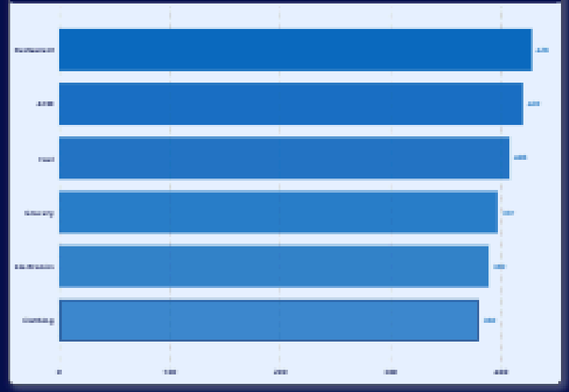
**MERCHANT CATEGORY**

| | | |
|---|---|---|
| Select all | Electronics | Restaurant |
| ATM | Fuel | |
| Clothing | Grocery | |

**FRAUD DISTRIBUTION BY CUSTOMER LOVATION**



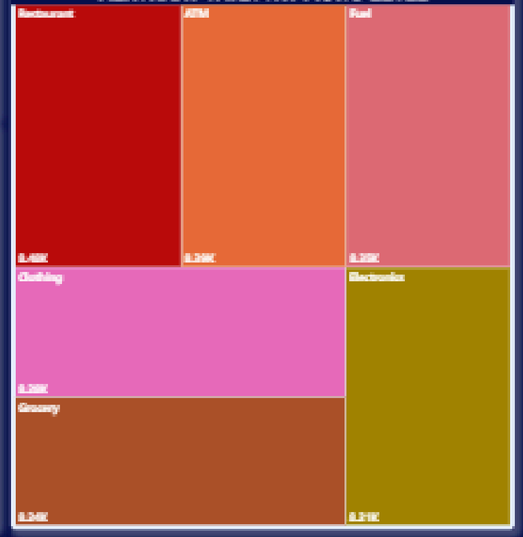

**FRAUD TRANSACTION BY MERCHANT CATEGORY**



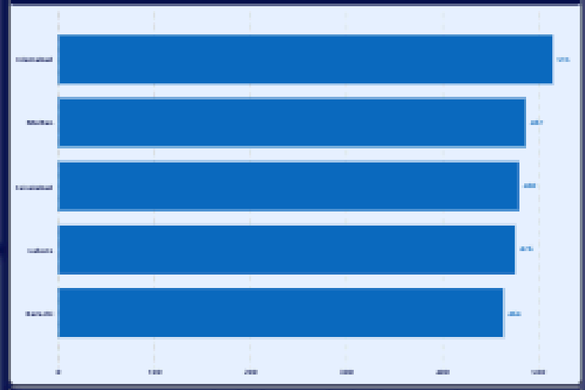
38K
Total_Customers

38K
Total Merchants

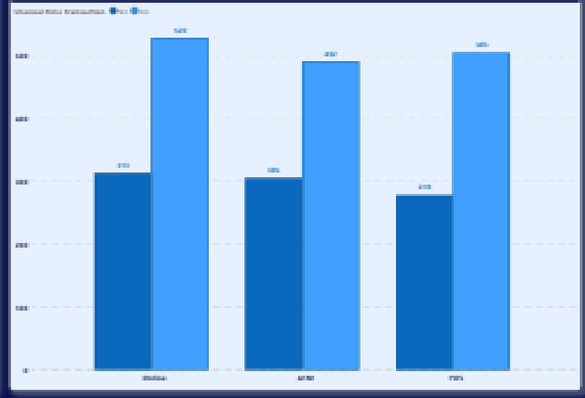
**MERCHANT CATEGORY FRAUD LEVEL**




**FRAUD TRANSACTION BY CUSTOMER LOCATION**



**FRAUD ACROSS TRANSACTION TYPES**



# 08 CONCLUSION

The Project successfully analyzed financial transaction data using Microsoft Excel and Power BI to identify fraud patterns and trends. The data was cleaned, transformed, and visualized through interactive dashboards, enabling analysis based on customer location, merchant category, transaction type, date and fraud status. The use of KPIs, charts, maps, and slicers improved the understanding of transaction behavior and supported effective decision-making. Overall, the project demonstrates how business intelligence tools can transform raw financial data into meaningful insights for fraud monitoring and analysis.

**FRAUD OCCURANCE by FAILED TRANSA**

| nt of Transaction_ID | Fraud_Label | |
|---|---|---|
| ed _Trans | Fraud | Normal |
| | 803 | 16011 |
| | 847 | 15841 |
| | 772 | 15675 |
| nd Total | **2422** | **47527** |

**FRAUD ANALYSIS FOR NEW MERCHANT vs**

| nt of Transaction_ID | Fraud_Label | |
|---|---|---|
| v_Merchant | Fraud | Normal |
| | 1197 | 23717 |
| | 1225 | 23810 |
| nd Total | **2422** | **47527** |

# THANK YOU

Let’s turn data into growth